

STAT332: Assignment 2 Question 5

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2. I made a small experiment to see how long it takes to boil (in minutes) a given amount of water on the four different burners of my stove (1 = right-back, 2 = right-front, 3 = left-back, 4 = left-front). Since salt affects the boiling point of water, I ran the experiment with 0, 2, 4 and 6 teaspoons of salt added to the water. Hence, my experiment two factors (4 levels each). In addition, I repeated the experiment 3 times for each of the 16 burner/salt combinations; yielding a total of 48 observations.

(a) Produce the ANOVA table.

```
water <- read.table("data/water.txt", header = FALSE)
colnames(water) <- c("burner", "salt", "time")
water$burner <- as.factor(water$burner)
water$salt <- as.factor(water$salt)
```

```
water.aov <- aov(time ~ burner * salt, data = water)
aov_tab <- summary(water.aov)
aov_tab
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## burner      3 110.50   36.83   73.667 1.86e-14 ***
## salt        3   6.83    2.28    4.556 0.00909 **
## burner:salt  9   3.33    0.37    0.741 0.66931
## Residuals   32  16.00    0.50
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(b) Perform the usual F tests in a two-way ANOVA. What are your conclusions on the effects of salt, burner and their interaction? Based on the ANOVA table, we can assess the p-value of each predictor as well as their interaction. If p-value > 0.05 , then we can reject the null hypothesis that the predictor is insignificant.

Since the p-value for burner is much smaller than 0.05, we conclude that there is very strong evidence that there is a significant effect on boiling time. More formally, we reject the null hypothesis that burner is an insignificant predictor.

The p-value for salt is smaller than 0.05, so we conclude that there is strong evidence that salt has a significant effect on boiling time. Formally, we reject the null hypothesis that salt is an insignificant predictor.

Contrarily, the p-value for the interaction between salt and burner is much greater than 0.05, so we conclude that there is very strong evidence that the effect of salt does not depend on the burner. We say that there is no interaction between the burner used and the amount of salt added to the water, hence the effect of salt is consistent across all burners.

(c) Using pairwise comparisons, which burner or burners is/are the fastest at boiling water? (Make sure that you clearly justify why only this/those burner(s) is/are faster than the others)

```
TukeyHSD(water.aov, "burner", conf.level = 0.95)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = time ~ burner * salt, data = water)
##
## $burner
##      diff      lwr      upr    p adj
## 2-1 -2.500000 -3.2821253 -1.717875 0.0000000
## 3-1 -0.5833333 -1.3654587  0.198792 0.2016864
## 4-1  1.7500000  0.9678747  2.532125 0.0000052
## 3-2  1.9166667  1.1345413  2.698792 0.0000010
## 4-2  4.2500000  3.4678747  5.032125 0.0000000
## 4-3  2.3333333  1.5512080  3.115459 0.0000000
```

When analyzing these pairwise results, we see that there is a negative difference between burner 1 and 2, where burner 2 is faster than burner 1 by 2.5 minutes. Furthermore, burner 2 is faster than burner 3 and burner 4 so we conclude that burner is the fastest at boiling water. It has the smallest mean boiling time and is significantly faster than all other burners based on Tukey pairwise comparisons.

- (d) Use a contrast to test (at the 5% level) if the two right burners take the same time to boil water than the left ones. No need to make multiple comparison adjustment.

We want to test the null hypothesis that $(\mu_1 + \mu_2) - (\mu_3 + \mu_4) = 0$.

```
means <- tapply(water$time, water$burner, mean)
means
```

```
##      1      2      3      4
## 6.666667 4.166667 6.083333 8.416667
```

```
contrast <- (means[1] + means[2]) - (means[3] + means[4])
contrast
```

```
##      1
## -3.666667
```

```
MSE <- aov_tab[[1]]["Residuals", "Mean Sq"]
df <- aov_tab[[1]]["Residuals", "Df"]
```

```
n <- 12
SE <- sqrt(MSE * (1/n + 1/n + 1/n + 1/n))
```

```
t <- contrast / SE
t
```

```
##      1
## -8.981462
```

```
2 * pt(-abs(t), df) # p-value
```

```
##          1
## 2.932649e-10
```

Since the p-value is much less than 0.05, we conclude that there is a statistically significant difference between the average boiling times of the right burners (1 and 2) and the left burners (3 and 4). At the 5% significance level, we reject the null hypothesis that the right and left burners have the same average boiling time.

Additionally, since the estimated contrast between the right burners and the left burners is negative, we also conclude that the right burners have lower boiling times than the left burners.

- (e) Use a contrast to test (at the 5% level) the hypothesis that there is a linear trend in the time to boil water due to the level of salt. No need to make multiple comparison adjustment. We will test the null hypothesis that there is no linear trend (i.e. $H_0 : -3\mu_1 - \mu_2 + \mu_3 + 3\mu_4 = 0$)

```
salt_means <- tapply(water$time, water$salt, mean)
salt_means
```

```
##          0          2          4          6
## 6.666667 6.000000 6.750000 5.916667
```

```
lin_contrast <- (-3 * salt_means[1]) + (-1 * salt_means[2])
               + (1 * salt_means[3]) + (3 * salt_means[4])
```

```
##          4
## 24.5
```

```
salt_SE <- sqrt(MSE * ((-3)^2 + (-1)^2 + 1^2 + 3^2) / 12)
t <- lin_contrast / SE
t
```

```
##          0
## -63.68673
```

```
2 * pt(-abs(t), df)
```

```
##          0
## 2.791283e-35
```

At the 5% significance level, we reject the null hypothesis of no linear trend in boiling time across salt levels. There is very strong evidence of a linear relationship between salt and boiling time. Furthermore, the negative test statistic indicates boiling time decreases as the amount of salt increases.